
Exercise sheet 6

Abelian Categories

January 13, 2026

The following exercises will be discussed in class.

Exercise 1

Let $\mathcal{C}$ be an **Ab**-category. Let X, Y be two objects of $\mathcal{C}$.

- (i) Show that the product of X and Y exists *if and only if* their coproduct exists. Show moreover that in that case the product and the coproduct are isomorphic. We call the resulting object the biproduct.
- (ii) Show that the biproduct of X and Y is characterized by an object Z together with morphisms

$$X \begin{array}{c} \xrightarrow{\iota_X} \\ \xleftarrow{\pi_X} \end{array} Z \begin{array}{c} \xrightarrow{\pi_Y} \\ \xleftarrow{\iota_Y} \end{array} Y$$

that satisfy the following equalities

$$\iota_X \pi_X = id_X, \quad \iota_Y \pi_Y = id_Y, \quad \pi_X \iota_X + \pi_Y \iota_Y = id_Z.$$

Exercise 2

Let $\mathcal{C}$ be a pre-abelian category. Let $\phi : X \rightarrow Y$ be a morphism of $\mathcal{C}$.

- (i) Show that there exists a canonical morphism $\bar{\phi} : \text{coker } \ker \phi \rightarrow \ker \text{coker } \phi$ such that ϕ decomposes as

$$X \xrightarrow{\pi} \text{coker } \ker \phi \xrightarrow{\bar{\phi}} \ker \text{coker } \phi \xrightarrow{\iota} Y$$

where π is an epimorphism and ι a monomorphism.

- (ii) Show that $\bar{\phi}$ is an isomorphism for every ϕ if and only if $\mathcal{C}$ is an abelian category.

Homework The following exercises will not be discussed in class and can be handed in until 26/01/2026. Feel free to only hand in partial answers or to ask for more details by email.

Exercise 1

Let $\mathcal{C}$ be an abelian category and let P be an object in $\mathcal{C}$. Show that the following statements are equivalent.

- (a) P is a projective object.
- (b) $\text{Hom}_{\mathcal{C}}(P, -)$ is an exact functor.
- (c) Every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ splits.

Exercise 2

Let R be a ring, not necessarily commutative. We consider the abelian category of left R -modules and we denote it $R\text{-Mod}$. Let P be a left R -module. Show that the following statements are equivalent.

- (a) P is a projective module.
- (b) There exists a set I and an R -module Q such that $P \oplus Q \cong R^{(I)}$. Here $R^{(I)}$ denotes the free R -module over the set I .
- (c) There exists a set I , a collection $\{x_i\}_{i \in I}$ of elements of P and a collection $\{f_i\}_{i \in I}$ of elements of $\text{Hom}_R(P, R)$ such that for all $x \in P$
 - there exist finitely many $i \in I$ with $f_i(x) \neq 0$;
 - $x = \sum_{i \in I} f_i(x)x_i$.